

Aditya Kulkarni

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SUMMARY

Full stack and AI engineer with production experience shipping multi-model LLM pipelines, agentic workflow orchestration, and distributed systems. 394 commits and 7 production deployments at Isomer AI. MS CS at NC State (May 2026). IEEE-published. 6+ hackathon wins including HackNC24 Top 10.

EXPERIENCE

Product Engineer Intern	Isomer AI	May 2025 - Present
- Developed 10+ backend features across 7 production deployments for an AI-powered claims automation platform; 394 commits in TypeScript, Node.js, Express, PostgreSQL, Redis, Prisma, and Docker with microservices architecture, real-time event pipelines (Pusher), CI/CD via GitHub Actions, and Sentry observability. - Architected Temporal.io durable workflow orchestration and multi-model LLM routing across Claude, GPT-4o, and Gemini (8+ models) via MCP tooling; built document processing pipelines (PDF generation, email parsing via Microsoft Graph API, S3 file storage), RBAC, and Auth0 authentication with Playwright automation.		
Graduate Research Assistant	NC State University	Apr 2025 - Present
Designed and deployed a cloud-native genomic data platform with JBrowse 2 on Google Cloud Run, serving 500GB of strawberry and grape datasets to 100+ researchers worldwide; improved ML pipeline accuracy by 12% via hyperparameter tuning and time-series analysis.		
Software Developer Intern	Mphasis (HP Company)	Jun 2024 - Aug 2024
Built 5+ AI-driven automation workflows including a health claim processing tool using OCR, NLP, and ML classification models, improving accuracy by 25%; automated CI/CD deployments on AWS and GCP, integrated Grafana monitoring dashboards, and supported model validation and red-teaming.		
AI/ML Intern	Capgemini	Jun 2023 - Aug 2023
Built distributed data pipelines using Hadoop and Apache Spark for batch processing and Apache Kafka streaming for real-time ingestion, automating support ticket analysis and reducing processing time by 15% and end-to-end latency by 20%.		

PROJECTS

AlgoPulse -- Agentic AI Orchestration	Temporal + Kafka
Re-architected agentic AI orchestration from API polling to event-driven streaming; Temporal.io durable workflow state with fault-tolerant retries, activity checkpointing, and saga-pattern rollback; Apache Kafka for partitioned worker coordination; 2,000 async LLM inference req/sec with zero message loss and sub-100ms p95 latency. TypeScript, Node.js, React, Supabase Edge Functions, Temporal.io, Kafka	
LLM Eval Harness	AI/ML Ops
Vendor-neutral LLMops benchmark comparing Claude Sonnet 4.6, GPT-4o, Gemini 2.5 Pro on 2,000 frozen examples; tracked hallucination rate, factual consistency, p50/p95 latency, token throughput, and cost-per-call; paired bootstrap significance testing; DuckDB + Parquet columnar storage; Streamlit regression dashboard. Python, Anthropic/OpenAI/Google SDKs, DuckDB, Parquet, Streamlit, Docker	
VectorLift -- Semantic Search Engine	Search / ML
Neural IR pipeline on MS MARCO: BM25 sparse baseline, dense bi-encoder retrieval (sentence-transformers, in-batch negatives), cross-encoder re-ranker for precision boosting; Apache Kafka-driven real-time indexing into Elasticsearch and Qdrant vector DB; FastAPI serving with per-stage latency tracing, Prometheus metrics, Grafana dashboards; NDCG@10/MRR@10 evaluation. Python, PyTorch, Elasticsearch, Qdrant, Kafka, FastAPI, Prometheus, Grafana, Docker	
PulseRank -- Real-Time Leaderboard	Distributed Systems
CQRS real-time leaderboard: MySQL transactional outbox + idempotency keys for exactly-once delivery, MongoDB TTL projections, Redis sorted sets for sub-ms rank reads; SSE live push; Testcontainers concurrency tests, k6 benchmarks (10K concurrent users), OpenTelemetry distributed tracing, Prometheus + Grafana observability. TypeScript, NestJS, Next.js, MySQL, MongoDB, Redis, Docker, OpenTelemetry, GitHub Actions	

EDUCATION

North Carolina State University -- MS Computer Science	Aug 2024 - May 2026
Artificial Intelligence, Operating Systems, Data Science, Software Engineering, Algorithms	
Mumbai University -- BTech Computer Science CGPA: 9.32/10	Aug 2020 - May 2024
Machine Learning, Big Data Infrastructure, DS & Algorithms, Blockchain	

SKILLS & AWARDS

Languages: Python, TypeScript, Java, JavaScript, SQL, C | **AI/ML:** MCP, LangChain, Hugging Face, PyTorch, RAG, OpenAI/Anthropic/Google SDKs | **Backend:** Node.js, FastAPI, Spring Boot, Express, NestJS | **Infra:** AWS, GCP, Docker, Kubernetes, Kafka, Temporal.io, Terraform | **Databases:** PostgreSQL, MySQL, MongoDB, Redis, Elasticsearch, Qdrant | **Observability:** Prometheus, Grafana, Sentry, OpenTelemetry

HackNC24 Top 10 (100+ teams, 48h) | **Sanjeevani Top 1%** (12 of 1,100+, 87% ML accuracy) | **IEEE Published** -- Mapify (500+ attendees) | **Google DSC Lead** (500+ students)